possible meaning for the “M” in M-theory) or—in deference to systematically naming their higher-dimensional cousins—*two-branes.* There are objects with three spatial dimensions called *three-branes.* And, although increasingly difficult to visualize, the analyses showed that there are also objects with p spatial dimensions, where p can be any whole number less than 10, known—with no derogation intended—as *p-branes.* Thus, strings are but one ingredient in string theory, not *the* ingredient.

These other ingredients escaped earlier theoretical investigation for much the same reason the tenth space dimension did: the approximate string equations proved too coarse to reveal them. In the theoretical contexts that string researchers had investigated mathematically, it turns out that all *p*-branes are significantly heavier than strings. And the more massive something is, the more energy is required to produce it. But a limitation of the approximate string equations—a limitation embedded in the equations and well known to all string theorists—is that they become less and less accurate when describing entities and processes involving more and more energy. At the extreme energies relevant for *p*-branes, the approximate equations lacked the precision to expose the branes lurking in the shadows, and that’s why decades passed without their presence being noticed in the mathematics. But with the various rephrasings and new approaches provided by the unified M-theoretic framework, researchers were able to skirt some of the previous technical obstacles, and there, in full mathematical view, they found a whole panoply of higher-dimensional ingredients.[3]

The revelation that there are other ingredients besides strings in string theory does not invalidate or make obsolete earlier work any more than the discovery of the tenth spatial dimension did. Research shows that if the higher-dimensional branes are much more massive than strings—as had been unknowingly assumed in previous studies—they have minimal impact on a wide range of theoretical calculations. But just as the tenth space dimension does not have to be much smaller than all others, so the higher-dimensional branes do not have to be much heavier. There are a variety of circumstances, still hypothetical, in which the mass of a higher-dimensional brane can be on a par with the lowest-mass string vibrational patterns, and in such cases the brane *does* have a significant impact on the resulting physics. For example, my own work with